

**WINTER 2025**

**SOIL HEALTH & CROP  
NUTRITION PLAN**

**FH Moses & Sons Pty Ltd**

**Prepared By : Sam Gulliford**

**TRADING NAME:** FH Moses & Sons Pty Ltd  
**ACCOUNT NUMBER:**  
**FARM NAME:** Valais  
**CONTACT:** James Moses  
**AREA (ha):** 107  
**DATE:** 27 May 2025

**ACCREDITED ADVISER:** Sam Gulliford  
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**LABORATORY:** Nutrient Advantage Lab

| <b>Paddock Name</b>             | <b>Bills</b>                                 | <b>Bills</b>                                 |
|---------------------------------|----------------------------------------------|----------------------------------------------|
| Area (ha)                       | 107                                          | 107                                          |
| <b>Target production (t/ha)</b> | <b>7.5</b>                                   | <b>0.0</b>                                   |
| Laboratory sample number        | 79678016                                     | 79671337                                     |
| Profile sampled (cm)            | 0-10                                         | 10-90                                        |
| Date of sampling                | 26 Mar 2025                                  | 26 Mar 2025                                  |
| Evaluation table                | Sorghum Grain Raingrown Liverpool Plains NSW | Sorghum Grain Raingrown Liverpool Plains NSW |
| Description                     | Bills                                        | Bills                                        |

## ANALYSIS RESULTS

| Paddock Name                             | Bills                | Bills               |
|------------------------------------------|----------------------|---------------------|
| Sample Depth (cm)                        | 0-10                 | 10-90               |
| Soil texture                             | Medium Clay          | Medium Clay         |
| Soil colour                              | Brown                |                     |
| Soil Colour (Munsell)                    | 2                    |                     |
| Colour (SM Code)                         | 2                    |                     |
| Soil Texture (Hand bolus)                | 6                    |                     |
| pH (1:5 H2O)                             | 7.1 - Satisfactory   | 8.6 - High          |
| pH (1:5 CaCl2)                           | 6.4 - Satisfactory   | 8 - High            |
| EC (1:5 H2O) dS/m                        | 0.12 - Satisfactory  | 0.19 - Satisfactory |
| EC (se) (dS/m)                           | 0.7 - Satisfactory   | 1.2 - Satisfactory  |
| EC (se) (dS/m) (Cladj)                   | 0.4 - Satisfactory   | 0.6 - Satisfactory  |
| Chloride (1:5 H2O) mg/kg                 | 10 - Satisfactory    | 10 - Satisfactory   |
| Organic carbon (Walkley Black) %         | 2.37 - High          |                     |
| Nitrate-N (Group) mg/kg                  | 18 - Marginal        | 4 - Unknown         |
| Ammonium nitrogen (KCl) mg/kg            | 3 - Sufficient       | 2 - Sufficient      |
| Phosphorus (Colwell) mg/kg               | 35 - Marginal        |                     |
| Phosphorus (BSES) mg/kg                  | 35                   |                     |
| Phosphorus Buffer Index (Colwell) (PBIC) | 160 - Sufficient     |                     |
| Phosphorus Environmental Risk Index      | 0.2 - Satisfactory   |                     |
| Potassium (Amm-Acet.) cmol+/kg           | 1.1 - Sufficient     |                     |
| Potassium % of CEC                       | 2.7 - Sufficient     |                     |
| Sodium:Potassium Ratio                   | 0.1 - Satisfactory   |                     |
| Sulfate-S (KCl40) mg/kg                  | 4.9 - Sufficient     | 4.6 - Sufficient    |
| Calcium (Amm-Acet) cmol+/kg              | 26 - Sufficient      |                     |
| Calcium % of CEC                         | 63 - Sufficient      |                     |
| Magnesium (Amm-Acet.) cmol+/kg           | 14 - High            |                     |
| Magnesium % cations                      | 33.9 - High          | -                   |
| Calcium:magnesium ratio                  | 1.9                  |                     |
| Calcium:Magnesium Ratio (cmol+/kg)       | 1.9                  |                     |
| Sodium (Amm-Acet.) cmol+/kg              | 0.14 - Sufficient    |                     |
| Sodium % cations                         |                      | -                   |
| Exch. sodium %                           | 0.3 - Satisfactory   |                     |
| Electrochemical Stability Index          | 0.353 - Satisfactory | -                   |
| Aluminium (KCl) cmol+/kg                 | 0.1 - Sufficient     |                     |
| Aluminium Saturation %                   | 1 - Sufficient       |                     |
| eCEC cmol+/kg                            | 41.2 - Satisfactory  | -                   |
| Copper (DTPA) mg/kg                      | 1.9 - Sufficient     |                     |
| Zinc (DTPA) mg/kg                        | 0.7 - Marginal       |                     |
| Manganese (DTPA) mg/kg                   | 46 - Sufficient      |                     |
| Iron (DTPA) mg/kg                        | 32 - Sufficient      |                     |
| Boron (hot CaCl2) (mg/kg)                | 0.7 - Sufficient     |                     |

## NUTRIENT REQUIREMENTS

| Paddock Name             | Bills        | Bills       |
|--------------------------|--------------|-------------|
| Area represented (ha)    | 107          | 107         |
| Target production (t/ha) | 7.5          | 0           |
| Growing Season           | 2025 Winter  | 2025 Winter |
| NITROGEN (kg/ha)         | 19.0 (122.7) |             |

| PHOSPHORUS (kg/ha)                  | 5.0 (15.3)  |        |             |          |            |      |        |                   |
|-------------------------------------|-------------|--------|-------------|----------|------------|------|--------|-------------------|
| SULFUR (kg/ha)                      | (2.8)       |        |             |          |            |      |        |                   |
| ZINC (kg/ha)                        | 1.00 (0.70) |        |             |          |            |      |        |                   |
| SOIL HEALTH                         |             | Result | Low         | Marginal | Sufficient | High | Excess | Sufficiency Range |
| Soil Colour (Munsell)               | 0 - 10      | 2.0    |             |          |            |      |        |                   |
| pH (1:5 H2O)                        | 0 - 10      | 7.1    | <div></div> |          |            |      |        | 5.5 - 8.5         |
|                                     | 10 - 90     | 8.6    | <div></div> |          |            |      |        | 5.5 - 8.5         |
| pH (1:5 CaCl2)                      | 0 - 10      | 6.4    | <div></div> |          |            |      |        | 4.7 - 7.7         |
|                                     | 10 - 90     | 8.0    | <div></div> |          |            |      |        | 4.7 - 7.7         |
| EC (1:5 H2O) dS/m                   | 0 - 10      | 0.12   | <div></div> |          |            |      |        | 0.00 - 0.80       |
|                                     | 10 - 90     | 0.19   | <div></div> |          |            |      |        | 0.00 - 0.80       |
| EC (se) (dS/m)                      | 0 - 10      | 0.7    | <div></div> |          |            |      |        | 0.0 - 6.8         |
|                                     | 10 - 90     | 1.2    | <div></div> |          |            |      |        | 0.0 - 6.8         |
| EC (se) (dS/m) (Cladj)              | 0 - 10      | 0.4    | <div></div> |          |            |      |        | 0.0 - 6.8         |
|                                     | 10 - 90     | 0.6    | <div></div> |          |            |      |        | 0.0 - 6.8         |
| Chloride (1:5 H2O) mg/kg            | 0 - 10      | 10     | <div></div> |          |            |      |        | 0 - 395           |
|                                     | 10 - 90     | 10     | <div></div> |          |            |      |        | 0 - 395           |
| Organic carbon (Walkley Black) %    | 0 - 10      | 2.37   | <div></div> |          |            |      |        | 1.00 - 2.00       |
| Phosphorus Environmental Risk Index | 0 - 10      | 0.20   | <div></div> |          |            |      |        | 0.00 - 0.65       |
| Sodium:Potassium Ratio              | 0 - 10      | 0.1    | <div></div> |          |            |      |        | 0.0 - 5.0         |
| Magnesium % cations                 | 0 - 10      | 33.9   | <div></div> |          |            |      |        | 0.0 - 25.0        |
|                                     | 10 - 90     |        |             |          |            |      |        |                   |
| Calcium:magnesium ratio             | 0 - 10      | 1.9    |             |          |            |      |        |                   |
| Calcium:Magnesium Ratio (cmol+/kg)  | 0 - 10      | 1.9    |             |          |            |      |        |                   |
| Sodium % cations                    | 10 - 90     |        |             |          |            |      |        |                   |
| Exch. sodium %                      | 0 - 10      | 0.3    | <div></div> |          |            |      |        | 0.0 - 6.0         |
| Electrochemical Stability Index     | 0 - 10      | 0.353  | <div></div> |          |            |      |        | 0.050 - 10.000    |
|                                     | 10 - 90     |        |             |          |            |      |        |                   |
| Dispersion Index (Loveday/Pyle)     | 0 - 10      | 0      | <div></div> |          |            |      |        | 0 - 6             |
| eCEC cmol+/kg                       | 0 - 10      | 41.2   | <div></div> |          |            |      |        | 5.0 - 100.0       |
|                                     | 10 - 90     |        |             |          |            |      |        |                   |

| <b>NUTRIENT</b>                          |         | <b>Result</b> | <b>Low</b> | <b>Marginal</b> | <b>Sufficient</b> | <b>High</b> | <b>Excess</b> | <b>Sufficiency Range</b> |
|------------------------------------------|---------|---------------|------------|-----------------|-------------------|-------------|---------------|--------------------------|
| Colour (SM Code)                         | 0 - 10  | 2.0           |            |                 |                   |             |               |                          |
| Soil Texture (Hand bolus)                | 0 - 10  | 6.0           |            |                 |                   |             |               |                          |
| Nitrate-N (Group) mg/kg                  | 0 - 10  | 18            |            |                 |                   |             |               | 30 - 50                  |
|                                          | 10 - 90 | 4             |            |                 |                   |             |               | 30 - 50                  |
| Ammonium nitrogen (KCl) mg/kg            | 0 - 10  | 3             |            |                 |                   |             |               | 0 - 5                    |
|                                          | 10 - 90 | 2             |            |                 |                   |             |               | 0 - 5                    |
| Phosphorus (Colwell) mg/kg               | 0 - 10  | 35            |            |                 |                   |             |               | 40 - 200                 |
| Phosphorus (BSES) mg/kg                  | 0 - 10  | 35            |            |                 |                   |             |               |                          |
| Phosphorus Buffer Index (Colwell) (PBIC) | 0 - 10  | 160           |            |                 |                   |             |               | 15 - 280                 |
| Potassium (Amm-Acet.) cmol+/kg           | 0 - 10  | 1.10          |            |                 |                   |             |               | 0.40 - 10.00             |
| Potassium % of CEC                       | 0 - 10  | 2.7           |            |                 |                   |             |               | 1.0 - 10.0               |
| Sulfate-S (KCl40) mg/kg                  | 0 - 10  | 4.9           |            |                 |                   |             |               | 4.0 - 50.0               |
|                                          | 10 - 90 | 4.6           |            |                 |                   |             |               | 4.0 - 50.0               |
| Calcium (Amm-Acet) cmol+/kg              | 0 - 10  | 26.0          |            |                 |                   |             |               | 1.0 - 100.0              |
| Calcium % of CEC                         | 0 - 10  | 63.0          |            |                 |                   |             |               | 55.0 - 90.0              |
| Magnesium (Amm-Acet.) cmol+/kg           | 0 - 10  | 14.0          |            |                 |                   |             |               | 1.6 - 10.0               |
| Sodium (Amm-Acet.) cmol+/kg              | 0 - 10  | 0.14          |            |                 |                   |             |               | 0.00 - 3.00              |
| Aluminium (KCl) cmol+/kg                 | 0 - 10  | 0.1           |            |                 |                   |             |               | 0.0 - 0.5                |



|                           |        |      |             |             |
|---------------------------|--------|------|-------------|-------------|
| Aluminium Saturation %    | 0 - 10 | 1.0  | <div></div> | 0.0 - 5.0   |
| Copper (DTPA) mg/kg       | 0 - 10 | 1.90 | <div></div> | 1.00 - 5.00 |
| Zinc (DTPA) mg/kg         | 0 - 10 | 0.70 | <div></div> | 0.80 - 5.00 |
| Manganese (DTPA) mg/kg    | 0 - 10 | 46.0 | <div></div> | 4.0 - 50.0  |
| Iron (DTPA) mg/kg         | 0 - 10 | 32.0 | <div></div> | 2.0 - 100.0 |
| Boron (hot CaCl2) (mg/kg) | 0 - 10 | 0.7  | <div></div> | 0.5 - 8.0   |

RECOMMENDATION: Bills

| TIMING                                   | PRODUCT                                 | APPLICATION         | RATE      | QUANTITY |
|------------------------------------------|-----------------------------------------|---------------------|-----------|----------|
| Pre plant spread                         | Urea - Granular (Incitec Pivot) (kg/ha) | Broadcast           | 250 kg/ha | 26.8t    |
| At planting                              | Granulock Z (Incitec Pivot) (kg/ha)     | Airseeder / seeding | 70 kg/ha  | 7.5t     |
| NUTRIENT RATES IN PRODUCT RECOMMENDATION |                                         |                     |           |          |
| NITROGEN (kg/ha)                         |                                         |                     | 122.7     |          |
| PHOSPHORUS (kg/ha)                       |                                         |                     | 15.3      |          |
| SULFUR (kg/ha)                           |                                         |                     | 2.8       |          |
| ZINC (kg/ha)                             |                                         |                     | 0.70      |          |

### Grower Details:

**Grower Name:** FH Moses & Sons Pty Ltd  
**Paddock Name:** Bills  
**Address:** "Valais" Quirindi 2343  
**Phone:**

**Interpretation Date:** Tuesday, 27 May 2025  
**Interpreter Name:** Sam Gulliford  
**Interpreter Phone:**

### Evaluation Parameters:

### Sample Details:

**Sample Date:** Wednesday, 26 March 2025  
**Forecast Plant Date:** 30-Apr-2025  
**Forecast Harvest Date:** 15-Dec-2026  
**Test Date:** Wednesday, 9 April 2025

**District:** Quirindi NSW  
**Description:** Bills  
**Crop:** Sorghum - Grain  
**Soil Type:** Black vertosol  
**Growth Stage:**

### YIELD

### Output Summary - DEMAND:

|                                                  | <u>Low</u> | <u>Expected</u> | <u>High</u> |
|--------------------------------------------------|------------|-----------------|-------------|
| <b>Target Yield (t/ha):</b>                      | 4.8        | 6.0             | 7.2         |
| <b>Target Protein (%):</b>                       | 9.5        | 9.5             | 9.5         |
| <b>Grain Removal:</b>                            | 73 kg/ha   | 91 kg/ha        | 109 kg/ha   |
| <b>Estimated Nitrogen Transfer Efficiency %:</b> | 50.0%      | 50.0%           | 50.0%       |
| <b>Estimated Crop N Demand (kg/ha):</b>          | 146 kg/ha  | 182 kg/ha       | 219 kg/ha   |
| <b>Estimated N in Crop (kg/ha):</b>              |            |                 |             |
| <b>N Required For Rest of The Crop (kg/ha)</b>   | 146        | 182             | 219         |

### Output Summary - SUPPLY:

### Sample Profile Details:

| Barcode   | Depth From (cm) | Depth To (cm) | Bulk Density (t/m3) | Nitrate N (mg/kg) | Ammonium N (mg/kg) | Depth Weighting | Mineral N (kg/ha) | Organic Carbon (%) |
|-----------|-----------------|---------------|---------------------|-------------------|--------------------|-----------------|-------------------|--------------------|
| 070276837 | 0.00            | 10.00         | 1.20                | 18.00             | 3.40               | 1.00            | 22                | 2.37               |
| 070276836 | 10.00           | 90.00         | 1.30                | 3.50              | 2.20               | 1.00            | 36                |                    |

|                                     |                |                 |                 |
|-------------------------------------|----------------|-----------------|-----------------|
| <b>Soil Mineral N:</b>              | 58 kg/ha       | 58 kg/ha        | 58 kg/ha        |
| <b>Rotation Credits:</b>            | 0 kg/ha        | 0 kg/ha         | 0 kg/ha         |
| <b>Soil Mineralisation Credits:</b> | 105 kg/ha      | 105 kg/ha       | 105 kg/ha       |
| <b>Manure Credits:</b>              | 0 kg/ha        | 0 kg/ha         | 0 kg/ha         |
| <b>Estimated Total Available N:</b> | 163 kg/ha      | 163 kg/ha       | 163 kg/ha       |
| <b>Recommended Nitrogen Rate:</b>   | <b>0 kg/ha</b> | <b>19 kg/ha</b> | <b>55 kg/ha</b> |

### Comments:

[Comments for depth 0.00 to 10.00]

[Comments for depth 10.00 to 90.00]

### **IMPORTANT NOTE:**

The evaluation and recommendation reports provided here are a guide only, and depend on representative samples being analysed. Additionally environmental and managerial factors influence production, therefore as suppliers of this report, we do not accept any liability whatsoever arising from the application of recommendations, for any damage, loss or injury of any nature and the user takes these evaluations and recommendations on these terms. This recommendation is made in good faith, based on the best technical information available.